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ENGLISH SUMMARY

One of the largest questions that astronomy attempts to answer is how our Solar System formed. Astronomers have spent centuries working on this problem by studying the heavens with progressively larger telescopes, trying to understand the formation of stars and planets, and whether it is possible for life to take root on a planet other than Earth.

Since the middle of the 20th century, an extraordinary amount of attention has been paid towards a small number of nearby star forming regions. These regions tend to contain a number of sparse groups of low-mass stars, forming in relative isolation to each other and their environment. Given their relative proximity and popularity, these have become the most well known, well understood sites of stars and planet formation in the Universe. An incredible wealth of information has been gained by studying these regions, propelling our understanding of star and planet formation dramatically. However, there is a limitation in studying these nearby regions. They almost certainly do not represent the environment that our Solar System formed in. Rather, multiple independent lines of evidence strongly suggest that our Solar System formed in a rich cluster, containing thousands of stars, alongside a number of massive stars.

Stellar clusters are characterised by high stellar densities, strong UV radiation fields, and dense clouds of interstellar gas and dust. It has been shown observationally that all of these environmental factors are capable of dramatically affecting how a star and its planets may form. Encounters between two stars can truncate protoplanetary disks. ISM material can fall onto disks after their formation, replenishing them with fresh material. UV photons can photoevaporate disks, leading to their rapid destruction. These environmental factors are weaker or entirely absent in the well studied nearby regions that we have built our foundational understanding of star and planet formation upon. The interaction between protoplanetary disks and the external environment remains poorly understood. The paradigm of star and planet formation still presupposes that there is no meaningful interaction, especially at later stages of evolution, despite a growing body of evidence to the contrary. It is therefore necessary to study more distant clusters, more representative of the Solar System's birthplace, in the same level of detail as those nearby regions. Doing so requires powerful telescopes and sensitive instrumentation, both of which are available in the era of JWST.

One of the fundamental processes that governs the birth of stars and planets is accretion. Accretion, in this context, is the transportation of material from the disk to the star. For the types of stars considered in this thesis, accretion is driven and moderated by the star's magnetic field. Magnetic field lines of the star extend from the stellar surface and reach into the disk itself. A magnetic pressure pushes the disk material back, creating a small gap between the disk and star at a location known as the truncation radius. Disk material at this location is drawn towards the star by these magnetic fields. The material

forms a dense, narrow column as it falls towards the stellar surface. The accretion column produces intense emission lines, which become brighter as more material is accreted. These emission lines can be measured, and their luminosity converted to a mass accretion rate, i.e. how quickly the star is growing.

The accretion rate onto the star sets constraints on how long a protoplanetary disk survives, and therefore on how long planets have to form. From studying nearby regions, it appears that disks survive around stars for a few million years. Accretion also influences the disk temperature, chemistry, structure and morphology, all of which impact planet formation. Is accretion a process that can be affected by the external environment? Accretion takes place very close to the star. At these small spatial scales, it would appear that the central star's influence should be much more important than the external environment. However, observations using the Hubble Space Telescope indicated that stars in clusters show systematically higher accretion rates for longer durations compared to stars in nearby regions. These observations were based solely on photometric measurements rather than spectroscopy, a necessary compromise due to the great distances towards these clusters. Photometry provides a feasible way of measuring the accretion rate of distant individual stars using narrow band filters centred on hydrogen lines such as H_α or Pa_β . However, photometry provides significantly weaker constraints on the stellar properties, leading to greater uncertainties in the ultimate accretion rates. High quality spectroscopic observations would be needed to confirm these unusually high accretion rates.

This thesis

The central theme of this thesis is accretion. The ultimate goal of the project was to utilise early JWST NIRSpec observations consisting of high quality spectroscopy of 100 stars in the Galactic starburst cluster NGC 3603. Photometric observations indicated that stars as old as 30 million years still show strong signatures of accretion in this cluster. With the resolution and sensitivity of JWST, it would now be possible to finally interrogate this finding spectroscopically. Due to the richness and quality of the obtained observations, several other projects were undertaken, all of which serve the overarching theme of stellar accretion in clusters.

In **Chapter 2** the wealth of nebular emission spectra that were obtained with NIRSpec were used in order to investigate the extinction properties of NGC 3603. The best fitting value of $R(V)$ was determined for each nebular spectra by fitting the decrement of Brackett emission lines. A value of 4.8 ± 1.06 was found for the cluster as a whole. This value of $R(V)$ was used in all subsequent papers when correcting the stellar spectra for extinction.

In **Chapter 3** the accretion luminosities of the young stars in our sample were calculated based on the line luminosity of Br_γ . This allowed, for

the first time, the calibration of both Pa_α and Br_β to measure the accretion luminosity. Both hydrogen lines are intrinsically strong, and suffer less from extinction compared to optical lines. This new calibration allows for the efficient measurement of the accretion luminosity for distant and embedded sources.

In **Chapter 4** a sample of intermediate mass and Herbig Ae stars from NGC 3603 were analysed in detail in order to constrain the origin of their hydrogen emission lines. Although it is known that both accretion and disk winds can contribute towards the line emission from young stars, it is generally unknown which of the two dominates for a given source. For Herbig Ae stars in particular, it remains uncertain whether magnetospheric accretion occurs at all. Two independent techniques were employed, exploiting the kinematics of the lines in order to reveal their origin. Distinct kinematic patterns were revealed in this analysis which can only be explained by magnetospheric accretion. This analysis provides a new method to constrain the origin of emission lines from spatially unresolved observations, and offers new evidence for magnetospheric accretion for Herbig Ae stars.

In **Chapter 5** the accretion properties of the young stars in NGC 3603 were analysed. The stellar properties were determined with a Markov chain Monte Carlo exploration, constraining the stellar temperature and luminosity. By fitting isochrones and evolutionary models, the mass, radius, and age of each star was determined. Their accretion rates were measured via their Br_γ line luminosities. In a number of cases our recent calibration of Pa_α was used to determine the accretion luminosity for sources where Br_γ could not be measured. The stars in our sample display systematically higher accretion rates compared to similar stars found in nearby regions, in agreement with previous results based on photometry. One quarter of the sample exhibit high accretion rates at old ages, with 12/42 stars consistent with ages of $>10\text{Myrs}$, and accretion rates of $\sim 10^{-8} \text{ M}_\odot \text{ yr}^{-1}$. Sources associated with high levels of nebular H_2 emission also exhibit accretion rates higher than expected given their age and mass. This relationship hints at a possible influence between the external environment, in terms of the interstellar gas density, and the stellar accretion rate.

NEDERLANDSE SAMENVATTING

Een van de grootste vragen die astronomie probeert te beantwoorden, is hoe ons zonnestelsel is gevormd. Astronomen werken al eeuwen aan dit probleem door de hemel te bestuderen met steeds grotere telescopen, in een poging te begrijpen hoe sterren en planeten gevormd worden, en of het mogelijk is dat leven zich kan ontwikkelen op een andere planeet dan de aarde.

Sinds het midden van de 20e eeuw is er buitengewoon veel aandacht besteed aan een kleine aantal nabije stervormingsgebieden. Deze gebieden bevatten vaak maar een paar schaarse verdeelde groepen van sterren met lage massa, die zich relatief geïsoleerd van elkaar en van hun omgeving vormen. Gezien hun relatieve nabijheid en populariteit zijn dit de bekendste en best begrepen locaties van sterren- en planeetvorming in het heelal geworden. Door het bestuderen van deze gebieden is een ongelooflijke hoeveelheid informatie verkregen, wat ons begrip van de vorming van sterren en planeten aanzienlijk heeft verdiept. Er is echter een beperking verbonden aan het bestuderen van deze nabije gebieden. Ze vertegenwoordigen vrijwel zeker niet de omgeving waarin ons zonnestelsel is gevormd. Verschillende onafhankelijke aanwijzingen wijzen er juist sterk op dat ons zonnestelsel is ontstaan in een rijk cluster met duizenden sterren, samen met een aantal zware sterren.

Sterrenhopen worden gekarakteriseerd door een hoge sterdichtheid, sterke UV-stralingsvelden en dichte wolken van interstellair gas en stof. Waarnemingen hebben aangetoond dat al deze omgevingsfactoren een drastische invloed kunnen hebben op hoe een ster en haar planeten gevormd worden. Nauwe ontmoetingen tussen sterren kunnen hun schijven afsnijden. Materiaal uit het interstellaire medium (ISM) kan na de vorming op de schijven vallen en ze aanvullen met vers materiaal. UV-straling kunnen schijven foto-evaporeren, wat leidt tot hun snelle vernietiging. Deze omgevingsfactoren zijn zwakker of zelfs volledig afwezig in de goed bestudeerde nabije gebieden waarop ons fundamentele begrip van sterren- en planeetvorming is gebaseerd. De interactie tussen protoplanetaire schijven en de externe omgeving is nog steeds slecht begrepen. Het heersende paradigma van sterren- en planeetvorming gaat er nog steeds van uit dat er geen betekenisvolle interactie plaatsvindt, ondanks een groeiend aantal aanwijzingen van het tegendeel. Het is daarom noodzakelijk om clusters, die beter representatief zijn voor de geboorteomgeving van ons zonnestelsel, met hetzelfde detailniveau te bestuderen als de nabije regio's. Dat vereist krachtige telescopen en gevoelige instrumentatie — beide zijn beschikbaar in het tijdperk van JWST.

Een van de fundamentele processen die de geboorte van sterren en planeten bepaalt, is accretie. Accretie betekent in deze context het transport van materiaal van de schijf naar de ster. Voor de typen sterren die in deze thesis worden behandeld, wordt accretie aangedreven en gereguleerd door het magnetisch veld van de ster. Magnetische veldlijnen van de ster strekken zich uit vanaf het steroppervlak tot in de schijf zelf. Een magnetische druk duwt het schijfmateriaal terug, waardoor een kleine kloof ontstaat tussen de schijf en de

ster, op een locatie die bekendstaat als de truncatieradius. Schijfmateriaal op deze locatie wordt door de magnetische velden naar de ster toe getrokken. Het materiaal vormt een dichte, smalle kolom terwijl het naar het oppervlak van de ster valt. Deze accretiekolom produceert intense emissielijnen, die helderder worden naarmate er meer materiaal wordt geaccreteerd. Deze emissielijnen kunnen worden gemeten, en hun luminositeit kan worden omgezet naar een massa-accretiesnelheid — oftewel hoe snel de ster groeit.

De accretiesnelheid op de ster bepaalt hoe lang een protoplanetaire schijf kan overleven, en daarmee hoe lang planeten de tijd hebben om zich te vormen. Uit studies van nabije regio's blijkt dat schijven enkele miljoenen jaren overleven. Accretie beïnvloedt ook de temperatuur, chemie, structuur en morfologie van de schijf, wat allemaal impact heeft op de planeetvorming. Is accretie een proces dat beïnvloed kan worden door de externe omgeving? Accretie vindt zeer dicht bij de ster plaats. Op deze kleine ruimtelijke schaal lijkt de invloed van de centrale ster veel belangrijker dan die van de externe omgeving. Echter, waarnemingen met de Hubble Space Telescope toonden aan dat sterren in clusters systematisch hogere accretiesnelheden laten zien gedurende langere tijd dan sterren in nabijgelegen regio's. Deze waarnemingen waren uitsluitend gebaseerd op fotometrische metingen en niet op spectroscopie, een noodzakelijke compromis vanwege de grote afstanden naar deze clusters. Fotometrie biedt een haalbare manier om de accretiesnelheid van verre individuele sterren te meten met behulp van smalbandfilters die gericht zijn op waterstoflijnen zoals H_α of Pa_β . Fotometrie levert echter aanzienlijk zwakkere beperkingen op de stereieigenschappen, wat leidt tot grotere onzekerheden in de uiteindelijke accretiesnelheden. Hoogwaardige spectroscopische waarnemingen zijn nodig om deze ongewoon hoge accretiesnelheden te bevestigen.

Dit proefschrift

Het centrale thema van deze scriptie is accretie. Het uiteindelijke doel van dit project was om vroege JWST NIRSpec-waarnemingen te benutten, bestaande uit hoogwaardige spectroscopie van 100 sterren in de Galactische stervormings-cluster NGC 3603. Fotometrische waarnemingen gaven aan dat sterren van wel 30 miljoen jaar oud nog steeds sterke accretiesignalen vertonen in deze cluster. Met de resolutie en gevoeligheid van JWST is het nu mogelijk om deze bevinding spectroscopisch te onderzoeken. Vanwege de rijkdom en kwaliteit van de verkregen waarnemingen zijn er meerdere aanvullende projecten uitgevoerd, die allen bijdragen aan het overkoepelende thema van stellaire accretie in clusters.

Hoofdstuk 2 gebruikte de grote hoeveelheid nevelmissiespectra die met NIRSpec zijn verkregen om de extinctie-eigenschappen van NGC 3603 te bestuderen. De best passende waarde van $R(V)$ werd voor elk nevelspectrum bepaald door de decrement van Brackett-emissielijnen te fitten. Voor de cluster als geheel werd een waarde van $4,8 \pm 1,06$ gevonden. Deze waarde van

$R(V)$ werd in alle volgende artikelen gebruikt bij het corrigeren van de sterrenspectra voor extinctie.

Hoofdstuk 3 berekende de accretieluminositeiten van de jonge sterren in onze steekproef op basis van de lijnluminositeit van Br_γ . Dit maakte het voor het eerst mogelijk om zowel Pa_α als Br_β te kalibreren voor het meten van de accretieluminositeit. Beide waterstoflijnen zijn intrinsiek sterk en ondervinden minder extinctie dan optische lijnen. Deze nieuwe kalibratie maakt een efficiënte meting van de accretieluminositeit mogelijk voor verre en ingebedde bronnen.

Hoofdstuk 4 analyseerde een steekproef van sterren met intermediaire massa en Herbig Ae-sterren uit NGC 3603 in detail om de oorsprong van hun waterstofemissielijnen te bepalen. Hoewel bekend is dat zowel accretie als winden kunnen bijdragen aan de lijnemissie van jonge sterren, is meestal onduidelijk welke van de twee dominant is voor een specifieke bron. Vooral voor Herbig Ae-sterren blijft onzeker of magnetosferische accretie überhaupt plaatsvindt. Twee onafhankelijke technieken werden toegepast, waarbij gebruik werd gemaakt van de kinematica van de lijnen om hun oorsprong te onthullen. In deze analyse werden duidelijke kinematische patronen gevonden die alleen verklaard kunnen worden door magnetosferische accretie. Deze analyse biedt een nieuwe methode om de oorsprong van emissielijnen uit ruimtelijk niet-gescheiden waarnemingen te bepalen en levert nieuw bewijs voor magnetosferische accretie bij Herbig Ae-sterren.

Hoofdstuk 5 analyseerde de accretie-eigenschappen van de jonge sterren in NGC 3603. De stereigenschappen werden bepaald met een Markov Chain Monte Carlo-verkenning, waarbij de ster temperatuur en -luminositeit werden beperkt. Door isochronen en evolutionaire modellen te fitten, werden massa, straal en leeftijd van elke ster bepaald. De accretiesnelheden werden gemeten via de Br_γ lijnluminositeiten. In een aantal gevallen werd onze recente kalibratie van Pa_α gebruikt om de accretieluminositeit te bepalen voor bronnen waarbij Br_γ niet gemeten kon worden. De sterren in onze steekproef tonen systematisch hogere accretiesnelheden dan vergelijkbare sterren uit lage-massa stervormingsgebieden, in overeenstemming met eerdere resultaten gebaseerd op fotometrie. Een kwart van de steekproef vertoont hoge accretiesnelheden op oudere leeftijden, waarbij 12 van de 42 sterren een leeftijd >10 miljoen jaar hebben en accretiesnelheden van ongeveer $10^{-8} M_\odot$ per jaar. Bronnen geassocieerd met hoge niveaus van nevelige H_2 -emissie vertonen eveneens hogere accretiesnelheden dan verwacht op basis van hun leeftijd en massa. Deze relatie wijst op een mogelijke invloed van de externe omgeving, in termen van de dichtheid van het interstellair gas, op de stellaire accretiesnelheid.

PUBLICATIONS

First author papers published in peer-reviewed journals

Externally irradiated young stars in NGC 3603. A JWST NIRSpec catalogue of pre-main-sequence stars in a massive star formation region

Ciarán Rogers, Guido De Marchi, and Bernhard Brandl

Published in *Astronomy & Astrophysics*

Volume 698, June 2025

Kinematic evidence of magnetospheric accretion for Herbig Ae stars with JWST NIRSpec

Ciarán Rogers, Bernhard Brandl, and Guido De Marchi

Published in *Astronomy & Astrophysics*

Volume 698, June 2025

Spectral characterisation of the extinction properties of NGC 3603 using JWST NIRSpec

Ciarán Rogers, Bernhard Brandl, and Guido De Marchi

Published in *Astronomy & Astrophysics*

Volume 688, August 2024

Determining stellar accretion rates from Pa_α and Br_β emission lines with JWST NIRSpec-Accretion of pre-main-sequence stars in NGC 3603

Ciarán Rogers, Guido De Marchi, and Bernhard Brandl

Published in *Astronomy & Astrophysics*

Volume 684, April 2024

First author papers published in non-peer-reviewed journals

Quantifying the contamination from nebular emission in NIRSpec spectra of massive star forming regions.

Ciarán Rogers, Guido De Marchi, Giovanna Giardino, Bernhard Brandl, Pierre Ferruit, and Bruno Rodriguez

Published in *Space Telescopes and Instrumentation 2022: Optical, Infrared, and Millimeter Wave* Volume 121803, August 2022

Co-author papers published in peer-reviewed journals

Protoplanetary Disks around Sun-like Stars Appear to Live Longer When the Metallicity is Low

Guido De Marchi, Giovanna Giardino, Katia Biazzo, Nino Panagia, Elena Sabbi, Tracy L. Beck, Massimo Robberto, Peter Zeidler, Olivia C. Jones, Margaret Meixner, Katja Fahrion, Nolan Habel, Conor Nally, Alec S. Hirschauer, David R. Soderblom, Omnarayani Nayak, Laura Lenkić, **Ciarán Rogers**, Bernhard Brandl, and Charles D. Keyes

Published in The Astrophysical Journal Volume 977, December 2024

Young Stellar Objects in NGC 346: A JWST NIRCам/MIRI Imaging Survey

Nolan Habel, Conor Nally, Laura Lenkić, Margaret Meixner, Guido De Marchi, Patrick J. Kavanagh, Katja Fahrion, Omnarayani Nayak, Alec S. Hirschauer, Olivia C. Jones, Katia Biazzo, Bernhard R. Brandl, J. Jaspers, Klaus M. Pontoppidan, Massimo Robberto, **Ciarán Rogers**, E. Sabbi, B. A. Sargent, David R. Soderblom, and Peter Zeidler

Published in The Astrophysical Journal Volume 971, August 2024

The early evolution of young massive clusters II. The kinematic history of NGC 6618/M 17

Mitchel Stoop, Annelotte Derkink, Lex Kaper, Alex de Koter, **Ciarán Rogers**, Maria Claudia Ramírez-Tannus, Difeng Guo and Naira Azatyan

Published in Astronomy & Astrophysics Volume 681, January 2024

JWST/NIRCам detections of dusty subsolar-mass young stellar objects in the Small Magellanic Cloud

Olivia C. Jones, Conor Nally, Nolan Habel, Laura Lenkić, Katja Fahrion, Alec S. Hirschauer, Laurie E. U. Chu, Margaret Meixner, Guido De Marchi, Omnarayani Nayak, Massimo Robberto, Elena Sabbi, Peter Zeidler, Catarina Alves de Oliveira, Tracy Beck, Katia Biazzo, Bernhard Brandl, Giovanna Giardino, Teresa Jerabkova, Charles Keyes, James Muzerolle, Nino Panagia, Klaus Pontoppidan, **Ciarán Rogers**, B. A. Sargent and David Soderblom

Published in Nature Astronomy Volume 7, April 2023

CURRICULUM VITAE

On the 25th of July 1995, I was born to two Irish parents, Ann and Rob, living in Manchester, England at the time. My sister Niamh, two years older than me, was naturally less than pleased with this development. At the turn of the millennium, my parents came to their senses and moved the family back to Dublin, Ireland, just in time for New Years Eve. I attended St. Fiachras primary school in Beaumont from 2000 to 2008. I had average maths skills and terrible handwriting, traits that I have carried with me to today.

Just after turning 13, I began secondary school at Ard Scoil Rís in Marino. I can recall in my second year, during science class, a student's book falling from their desk to the floor. As they began to pick up the book, my teacher, Mr O'Connor, stopped them. He explained that their book was trying to get to the centre of the earth, but the floor was stopping it. So long as the book lay where it was, it could not fall any more, so better to just leave it alone. I had never considered the force of gravity in quite those terms before. I believe that this was the moment that I became interested in physics. I graduated from Ard Scoil Rís in 2013 after completing my Leaving Certificate exams, obtaining 470 out of 625 possible points, enough to secure my place in the General Science degree at Dublin City University, more commonly referred to as DCU.

I was open to pursuing either physics or biology when I began my bachelor's degree. The practical lab sessions thoroughly convinced me that I was not cut out to deal with petri dishes, and I gladly opted to specialise in Physics and Astronomy for the remainder of my degree. I graduated in the Autumn of 2017 with first class honours.

I was not sure what I wanted to do, or be, following my bachelor's degree. I lived with two friends in Chicago in the summer of 2017, working in the old town as a waiter at the Italian restaurant Orso's. I spent the following year tutoring students in maths, physics and biology, and teaching science courses to young children on weekends at my former university, DCU.

My time was mostly occupied with playing Tony Hawks Underground on Playstation 2. Understandably, the novelty of having near infinite free time soon became a burden. More than anything, I wanted to move out and find my own place. I did what many twenty-somethings do when they are lacking direction, I started applying for Master's degrees. I thought I could kill two birds with one stone by applying outside of Ireland, where rent may be more affordable. Almost immediately I received an invitation from the University of Amsterdam to enrol in their Science for Energy and Sustainability Master's programme. I had felt a growing disconnect to astronomy in this time, and sought a more practical application of my skills. However, with virtually no savings and therefore no feasible way of affording a two year Master's in The Netherlands' most expensive city, I declined the invitation, determined to re-apply the following year with a healthier bank balance.

I gained employment working for Irish Water, the state water utility com-

pany. Hired initially as a graduate engineer, I quickly became useful as the only person in the office with any programming experience. I learned how to write in Visual Basic for Applications, or VBA, the language used to interact with Microsoft Office applications, like Excel. Informally, I became the office data engineer, automating and optimising the application process through which contractors apply for work through Irish Water. I can remember during lunch one afternoon, reading about the Event Horizon Telescope taking the first picture of a black hole. I realised that myself and astronomy hadn't actually broken up, I just needed some space. When applications for UvA opened again, I applied for the astronomy programme.

My partner Jessie and I moved from Dublin to Amsterdam in the late summer of 2019. We had managed to thread the needle, both enrolling in Master's programmes at UvA at the same time. We had found a beautiful apartment next to Vondel Park. We watched the first half of *All the President's Men* before falling asleep on our first night there. The first six months of living in Amsterdam were intense and exhilarating. Discovering Amsterdam, meeting new friends, doing my own laundry, it was an era of firsts. My degree was ultimately impacted by Covid 19, with the majority taking place strictly online. I finished my degree with the conclusion of my Master's research project, studying the variability of massive stars in M17, supervised by Dr. Annelotte Derkink and Prof. Lex Kaper. For the first time, I received a glimpse into what observational astronomy looked like. The reduction and analysis of those optical spectra from the William Herschel Telescope completely rewired my brain. I realised that I wasn't nearly ready to stop. Despite my continued protestations throughout my degree that I would not pursue a PhD, I found myself feverishly applying in the Autumn of 2020.

I was offered a position at Leiden University, working with Prof. Bernhard Brandl and Dr. Guido De Marchi. The research would consist of brand new JWST observations from NIRSpec and MIRI focusing on star formation in large clusters. This project was by far the one that I wanted most, and it goes without saying that I'm glad their first choice turned it down. I moved from Amsterdam to Leiden in Autumn of 2021. My time at Leiden has been blissful. I adore the city. I consider myself fortunate that my experience at the observatory has been overwhelmingly positive. The cohort of Master's students, PhD candidates and Post-docs makes it an extremely lively environment which I am going to dearly miss. My PhD research morphed and evolved organically over the four years, sliding from star formation, to protoplanetary disks and stellar accretion, to the impact of the external environment on star and planet formation. Bernhard and Guido were flexible and understanding with the direction my PhD took. They provided guidance when it was needed, while allowing me to make mistakes that would teach me important lessons. I was encouraged to submit a JWST Cycle 3 proposal following an exciting meeting with Dr. Olivier Berné at STScI in Baltimore. This program was accepted, and I view this as probably the most important milestone of my PhD.

It changed my perspective on what I was capable of achieving in astronomy, and provided me with enough confidence to apply for fellowships to pursue my own independent research following my PhD.

At the end of 2025 I will begin a position at STScI as an ESA research fellow. My Cycle 3 program will make up a large fraction of my research there. The inception of that program came about from a chance encounter at STScI two years earlier. This full circle moment brings me a lot of joy whenever I think about it.

ACKNOWLEDGEMENTS

I want to begin by thanking my two supervisors, Bernhard Brandl and Guido De Marchi, who I affectionately refer to as ‘the supes’ (though I don’t think they are aware of this). You gave me the opportunity to work in Leiden on eye-watering observations, more beautiful and rich and inspiring than I deserved. Your faith, encouragement and guidance have propelled me through an intense period of growth and change, something I have in common with my favourite young stars. I feel comfortable calling myself an astronomer at the end of this process, and I have you both to thank for that. Thank you Bernhard. Thank you Guido.

The sterrewacht at Leiden is a vast and immersive community of researchers and students. I felt immediately invited and welcomed to join that community. Within a couple of days of arriving in Leiden, before my contract had even begun, I found myself at a summertime borrel outside of Huygens enjoying crisp Amstels and Leffes with Sill Verberne, Joshua Butterworth, Roi Kugel, Nicco Veronesi, Andrew Allan and Rob Kavanagh. The countless subsequent borrels, PhD defence parties, ski-trips, stupid food challenges, as well as all of the conferences that I had the privilege of attending with my new friends, these experiences have been the bedrock of my social life in Leiden.

My fellow JWST observers, Ivana van Leeuwen, Lucie Rowland, Amy Louca, Thomas Herard, Darío González Picos, Logan Francis and Lukasz Tychoniec, thank you for providing sympathetic ears when I would complain about this fabulous observatory. I need to specifically acknowledge my two NIRSspec MSA sisters, Chloe Cheng and Martje Slob. Wandering into your office to ponder the mysteries of the MSA was genuinely some of the most rewarding collaboration I experienced in Leiden.

Osmar and Kevin, my amigos. You guys throw the best parties. Here’s to more living room karaoke.

My slightly more boyish doppelganger, Joey Braspenning. Bedankt voor jouw vriendschap. Ik kijk uit naar de volgende ski-trip, en natuurlijk ook de sauna. Tot ziens makker.

Probably the single most important influence in my life has been the work ethic of my parents, Ann and Rob. The only reason why any of this has happened is because you put me in the position for it to be possible. What feels like my natural tendency to push hard in my career, has been learned and assimilated from both of you. I really don’t know people who have more endurance and commitment than both of you, and these qualities are my greatest inheritance.

Finally, I need to address the single most important person during my time as a PhD student, as well as during all other times that I’ve known her, my partner, Jessica Whittle. We have gone through this entire Dutch odyssey together, completing our Master’s degrees at UvA, going stir-crazy during Covid, and fantasizing about the modest luxuries that we would afford once we both had jobs. Free-way cola was just as good as the real thing if I was

drinking it with you.

We celebrated ten years together in Leiden, realising that we had now spent more time in the Netherlands than we had in Dublin. We've both come to understand the Netherlands in the same way, harshly criticising obnoxious Albert Heijn advertisements, while being unabashedly in love with this country, our home.

Although I was surrounded by fellow astronomers, you were the person who was most capable of understanding and empathizing with my experience. The low points would not have been bearable without you, and the high points are only meaningful because you're there. You are the most encouraging and supportive person that I know. You have bolstered me with a confidence and a belief that has fundamentally changed who I am, and the kind of person I want to be. You have gone through an incredible period of growth yourself. As proud as I am for what I have done with your support, I am even more proud of your own achievements. Multiple times, you made both personal and professional accomplishments that mere months before seemed like ludicrous pipe dreams. Your ambition to get what you want, in a foreign language in a foreign place, requires bravery and self belief that few people have.

None of the last six years would have happened without you. Thank you for dealing with me (especially during fellowship applications). This year has been the most challenging and overwhelming that we've experienced. Who'd have thought achieving your dreams could be so stressful? You have taught me that the most important thing to do during times like these is to be kind to each other. The storm eventually passes, and what's left is the love, the bond, the friendship of the two people who weathered it together. Thanks babe.